

Fishing for Sustainability

Assignment Submission

Wed Mar 25, 2026

Name: _____

Each person should submit their own copy of this handout at the end of class for today's completion credit.

Round 1: Open Access

Data Table

Item	Value
Your catch	
Total class catch	
Fish remaining after harvest	
Fish after reproduction (doubled, max 30)	

Reflection

(1) Based on what happened, is this situation a **social dilemma**? Explain briefly.

(2) In one sentence, explain how Round 1 illustrates the **Tragedy of the Commons** (focus on individual incentives vs. group outcomes).

Round 2: Fixed Quota

Data Table

Item	Value
Agreed class quota	
Your catch	
Total class catch	
Fish remaining after harvest	
Fish after reproduction (doubled, max 30)	

Reflection

- (3) During the discussion, did the class choose a quota aimed at **maximizing long-term sustainability** or **maximizing short-term catch**?

- (4) How does this round illustrate challenges of **enforcement, trust, or cheating** in managing common-pool resources?

Round 3: Fixed Effort

Data Table

Item	Value
Agreed per-person limit	
Total class catch (limit $\times$ 8)	
Fish remaining after harvest	
Fish after reproduction (doubled, max 30)	

Reflection

- (5) Was it easier to comply with this rule compared to the quota in Round 2? Why or why not?

- (6) How does this round reduce (or fail to reduce) the **social dilemma** you experienced earlier?

Synthesis

Data Overview

Round	Strategy	Total.Catch	Fish.Remaining.After.Harvest	Fish.After.Reproduction
1	Open Access			
2	Fixed Quota			
3	Fixed Effort			

Synthesis Questions

- (7) Looking at your data, which round left the fish population **closest to the midpoint (~15 fish)** after harvest? Why does leaving the population near this level tend to produce the **highest sustainable yield over time**?

- (8) For each strategy, give **one advantage and one drawback** based on your experience:

- Open Access:

- Fixed Quota:

- Fixed Effort (Per-Person Limit):

- (9) Which round depended most on **trust** between individuals? Which round relied most on **enforcement**? Why are these factors so important in managing real-world common-pool resources?

(10) If you were managing a real fishery, which strategy (or combination of strategies) would you recommend?

(11) Name **two specific actions** you would implement in your strategy from the previous question (e.g., monitoring, enforcement, adaptive limits, stakeholder involvement).

(12) Give one example of another **common-pool resource** (e.g., groundwater, forests, wildlife, carbon emissions). Which management approach from this activity might work best there—and why?
